EECS 836: Machine Learning

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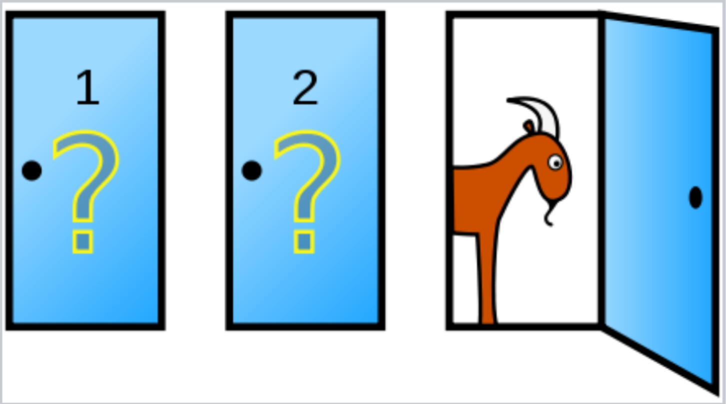
The Monty Hall problem

- Suppose you're on a game show, and you're given the choice of three doors.
  - Behind one door is a car; behind the others, goats.
  - You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat.
  - He then says to you, "Do you want to switch to door No. 2?"
- Is it to your advantage to switch your choice?
- A: Car is behind door 1; B: goat is behind door 3
- What is the probability of P(A|B)?



In search of a new car, the player picks a door, say 1. The game host then opens one of the other doors, say 3, to reveal a goat and offers to let the player pick door 2 instead of door 1.

https://youtu.be/i7OY7CZUdIY

Agenda

- Probability Basics
  - Probability and Bayes' Rules
- Probabilistic Learning
  - Maximum Likelihood Estimation (MLE)
  - Maximum A Posteriori (MAP) Estimation
- Naive Bayes Classifier

Probability of binary event

Binary random variable

- Let A denote a binary random variable
  - A represents an event with uncertainty
  - Whether A occurs has an associated probability
- Examples of binary random variables:
  - $A =$ I get head in a coin toss
  - $A =$ It rains today
- $P(A = True)$ or $P(A)$ indicates "the fraction of possible worlds in which A is true"
  - Accordingly, $P(A = False)$ or $P(\sim A)$ means the probability that event A does not occur

Axioms of probability

- All probabilities are between 0 and 1:

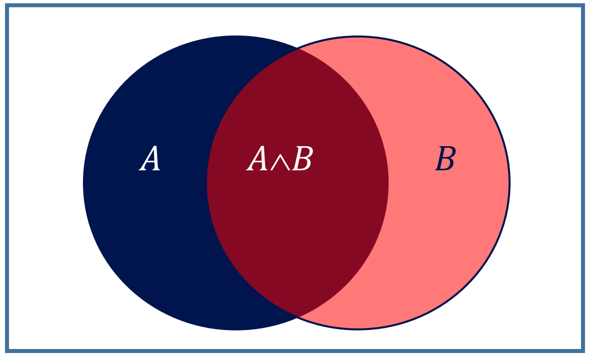
$$0 \leq P(A) \leq 1$$

- Valid propositions have probability 1; unsatisfiable propositions have probability 0:

$$P(True) = 1 ; \quad P(False) = 0$$

- The probability of a disjunction is given by:

$$P(A \vee B) = P(A) + P(B) - P(A \wedge B)$$



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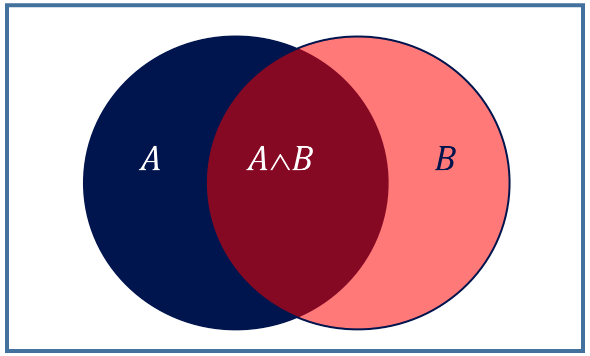

- The probability of a disjunction is given by:

$$P(A \vee B) = P(A) + P(B) - P(A \wedge B)$$

From these we can prove more properties:

$$P(\sim A) = 1 - P(A)$$

$$P(A) = P(A \wedge B) + P(A \wedge \sim B)$$

Probability of multi-valued event

- A is a *random variable with k unique values* if it can take on exactly one value out of $\{v_1, v_2, \ldots, v_k\}$
- Think about tossing a die – get a number 1~6
- Important properties
  - $P(A = v_i \wedge A = v_j) = 0 \;\; if \quad i \neq j$
  - $P(A = v_1 \vee A = v_2 \vee \ldots \vee A = v_k) = 1$

Multi-valued variable

$$1 = \sum_{i=1}^{k} P(A = v_i)$$

Probability of multi-valued event

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Multi-valued variable

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  - $P(A = v_1 \vee A = v_2 \vee \ldots \vee A = v_k) = 1$

$$1 = \sum_{i=1}^{k} P(A = v_i)$$

- Marginalization over *A*
  - $P(B) = P(B \wedge [A = v_1 \vee A = v_2 \vee \cdots \vee A = v_k])$

$$\mathrm{P}(B) = \sum_{i=1}^{k} P(B \wedge A = v_i)$$

Joint probability

- Considering **3** binary random variables cases
  - The joint probability is given by

	$alarm$		$\sim alarm$	
	$earthquake$	$\sim earthquake$	$earthquake$	$\sim earthquake$
$burglary$	0.01	0.08	0.001	0.009
$\sim burglary$	0.01	0.09	0.01	0.79

$$P(alarm) = \sum_{b,e} P(alarm \wedge burglary = b \wedge earthquake = e) = 0.01 + 0.08 + 0.01 + 0.09 = 0.19$$

$$P(burglary) = \sum_{a,e} P(alarm = a \wedge burglary \wedge earthquake = e) = 0.01 + 0.08 + 0.001 + 0.009 = 0.1$$

Joint probability

- Joint probability: matrix of combined probabilities of a set of variables
- Considering **2** binary random variables cases
  - The joint probability is given by

	$alarm$	$\sim alarm$
$burglary$	0.09	0.01
$\sim burglary$	0.1	0.8

Probability of burglary: $P(burglary) = 0.1$

by marginalization over $alarm$: $P(burglary) = P(burglary \wedge alarm) + P(burglary \wedge \sim alarm)$

Joint probability

- Considering **3** binary random variables cases
  - The joint probability is given by

	$alarm$		$\sim alarm$	
	$earthquake$	$\sim earthquake$	$earthquake$	$\sim earthquake$
$burglary$	0.01	0.08	0.001	0.009
$\sim burglary$	0.01	0.09	0.01	0.79

$$P(alarm) = \sum_{b,e} P(alarm \wedge burglary = b \wedge earthquake = e) = 0.01 + 0.08 + 0.01 + 0.09 = 0.19$$

Get joint probabilities

- For making a joint distribution of d variables
    1. Make a truth table listing all combinations of values of your variables (if there are d Boolean variables then the table will have 2^d rows).

Boolean variables A, B, C

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Get joint probabilities

- For making a joint distribution of d variables
  1. Make a truth table listing all combinations of values of your variables (if there are d Boolean variables then the table will have 2^d rows).
  2. For each combination of values, say how probable it is.

Boolean variables A, B, C

A	B	C	Prob
0	0	0	0.30
0	0	1	0.05
0	1	0	0.10
0	1	1	0.05
1	0	0	0.05
1	0	1	0.10
1	1	0	0.25
1	1	1	0.10

Get joint probabilities

- For making a joint distribution of d variables
  1. Make a truth table listing all combinations of values of your variables (if there are d Boolean variables then the table will have 2^d rows).
  2. For each combination of values, say how probable it is.
  3. Verify that all the probabilities sum up to 1.

Boolean variables A, B, C

A	B	C	Prob
0	0	0	0.30
0	0	1	0.05
0	1	0	0.10
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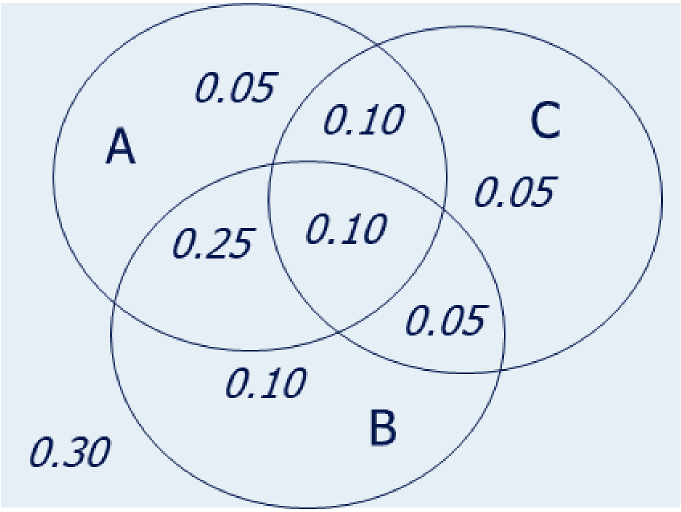
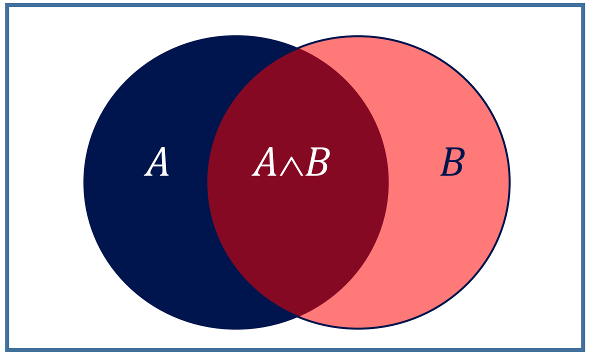
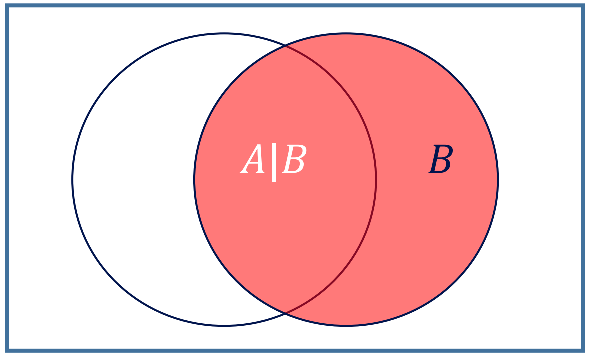


Conditional probability

- $P(A \mid B)$ = Fraction of worlds in which B is true that also have A true
- If we already know that B is true
  - The knowledge changes the probability of A
  - Because we know we are in a world where B is true





$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$
$$P(A \wedge B) = P(A|B) \times P(B)$$

Conditional probability

$P(alarm \wedge burglary)$ =

	$alarm$	$\sim alarm$
$burglary$	0.09	0.01
$\sim burglary$	0.1	0.8

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$

$$P(burglary \mid alarm) = P(burglary \wedge alarm) \,/\, P(alarm)$$
$$= 0.09 \,/\, 0.19 = 0.47$$

$$P(alarm \mid burglary) = P(burglary \wedge alarm) \,/\, P(burglary)$$
$$= 0.09/0.1 = 0.9$$

Conditional probability

$P(alarm \wedge burglary)$ =

	$alarm$	$\sim alarm$
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$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$

$$P(A \wedge B) = P(A|B) \times P(B)$$

$$P(burglary \mid alarm) = P(burglary \wedge alarm) \,/\, P(alarm) = 0.09 \,/\, 0.19 = 0.47$$

$$P(alarm \mid burglary) = P(burglary \wedge alarm) \,/\, P(burglary) = 0.09/0.1 = 0.9$$

$$P(burglary \wedge alarm) = P(burglary \mid alarm)\, P(alarm) = 0.47 * 0.19 = 0.09$$

Independence

- When two event do not affect each others' probabilities, we call them independent
- Formal definition:

$$A \perp\!\!\!\perp B \quad \leftrightarrow P(A \wedge B) = P(A) \times P(B)$$
$$\leftrightarrow P(A|B) = P(A)$$

Conditional independence

- Absolute independence of A and B:

$$A \perp\!\!\!\perp B \quad \leftrightarrow P(A \wedge B) = P(A) \times P(B)$$
$$\leftrightarrow P(A|B) = P(A)$$

- Conditional independence of A and B given C

$$\boxed{A \perp\!\!\!\perp B |\, \mathrm{C} \leftrightarrow P(A \wedge B\ |C) = P(A|C)\ \times P(B|C)}$$

- This lets us decompose the joint distribution:
  - $P(A, B, C)$ = [Always] $P(A|B, C)P(B|C)\, P(C)$ = $P(A, B|C)P(C)$
  - = [Independence] $P(A|C)\, P(B|C)\, P(C)$
  - Conditional independence is weaker than absolute independence, but still useful to decompose the full joint

Summary: the Monty Hall problem

- Suppose you're on a game show, and you're given the choice of three doors.
  - Behind one door is a car; behind the others, goats.
  - You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat.
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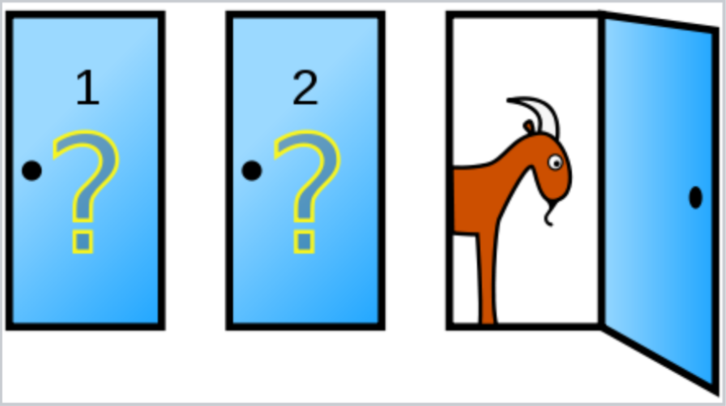

In search of a new car, the player picks a door, say 1. The game host then opens one of the other doors, say 3, to reveal a goat and offers to let the player pick door 2 instead of door 1.

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Basic probability

- Product Rule: $P(A,B) = P(A|B)P(B) = P(B|A)P(A)$
- If A and B are independent:
  - $P(A,B) = P(A)P(B); \quad P(A|B) = P(A), P(A|B,C) = P(A|C)$
- Sum Rule: $\mathrm{P}(\mathrm{A} \vee B) = P(A) + P(B) - P(A,B)$
- Total Probability:
  - If events $A_1, A_2, \ldots A_n$ are mutually exclusive: $A_i \wedge A_j = \Phi$, $\sum_i P(A_i) = 1$
  - $P(B) = \sum P(B, A_i) = \sum_i P(B|A_i)\, P(A_i)$
- Total Conditional Probability:
  - If events $A_1, A_2, \ldots A_n$ are mutually exclusive: $A_i \wedge A_j = \Phi$, $\sum_i P(A_i) = 1$
  - $P(B|C) = \sum P(B, A_i|C) = \sum_i P(B|A_i, C)\, P(A_i|C)$

Refer to Tutorial 3 – basics of probability for more details.

Bayes' Rule

- The most important formula in probabilistic machine learning

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Derivation:
$P(A \wedge B) = P(A|B) \times P(B)$
$P(B \wedge A) \ = P(B|A) \times P(A)$

Having them equal, we get

$$P(A|B) \times P(B) = P(B|A) \times P(A)$$

Bayes, Thomas (1763) An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, **53:370-418**

Bayes' Rule interpretation

- The most important formula in probabilistic machine learning

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

- Allow us to reason from evidence (data) to hypotheses (target/model)

$$P(\text{hypothesis}\ |\text{evidence}) = \frac{P(\text{evidence}\ |\text{hypothesis}) \times P(\text{hypothesis})}{P(\text{evidence})}$$

Bayes' Rule interpretation

- The most important formula in probabilistic machine learning

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

- Allow us to reason from evidence to hypotheses

Likelihood

Prior probability

$$P(\text{hypothesis}\ |\text{evidence}) = \frac{P(\text{evidence}\ |\text{hypothesis}) \times P(\text{hypothesis})}{P(\text{evidence})}$$

Posterior probability

Marginal probability

Basics of Bayesian learning

- Goal: find the best hypothesis from some space H of hypotheses, given the observed data (evidence) D.

- Define best to be: most probable hypothesis in H
- In order to do that, we need to assume a probability distribution over the class H.

- In addition, we need to know something about the relation between the data observed and the hypotheses (E.g., a coin problem.)
  - We will be Bayesian about other things, e.g., the parameters of the model

Bayes' Theorem

- Allow us to reason from evidence to hypotheses

$$P(h\,|D) = \frac{P(D\,|h) \times P(h)}{P(D)}$$

- $P(h|D)$ increases with $P(h)$ and with $P(D|h)$
- $P(h|D)$ decreases with $P(D)$

Basics of Bayesian learning

- Allow us to reason from **evidence (data)** to **hypotheses (parameters)**

$$P(h|D) = \frac{P(D|h) \times P(h)}{P(D)}$$

- $P(h)$ - the prior probability of a hypothesis h, reflecting background knowledge before data is observed.
- $P(D)$ - The marginal probability that this sample of data is observed. (No knowledge of the hypothesis)
- $P(D|h)$ – The likelihood of observing the sample D given the hypothesis h
- $P(h|D)$ - The posterior probability of h. The probability of hypothesis h given that D has been observed.

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  - Probability and Bayes' Rules
- Probabilistic Learning
  - Maximum Likelihood Estimation (MLE)
  - Maximum A Posteriori (MAP) Estimation
- Naive Bayes Classifier

Example of probabilistic learning

- Make a guess: Kansas City Chiefs' probability of winning at the end of upcoming season?

An example based on Shota Horii tutorial: link

Example of probabilistic learning

- Make a guess: Kansas City Chiefs' probability of winning at the end of upcoming season?
- Let's collect some evidence: in 2021 season, Kansas City won 12 games of 17 games in NFL
  - $\boxed{\text{The simplest guess would be 12/17=71\%, the best guess based on data}}$

This is Maximum Likelihood Estimation (MLE)!

Example of probabilistic learning

- Make a guess: Kansas City Chiefs' probability of winning at the end of current season?
- Let's collect some evidence: in 2022 season, Kansas City won 12 games of 17 games in NFL
  - The simplest guess would be 12/17=71%, the best guess based on **data**

This is Maximum Likelihood Estimation (MLE)!

  - Then we know that Kansas City's winning probability for the past few seasons were around 50%
    - Do you think our best guess is still 71%? Maybe a value between 50% and 71%?
    - Considering the **prior knowledge** and the **data** we make the best guess

This is Maximum A Posteriori (MAP)!

Example of probabilistic learning

- MLE and MAP are methods of estimating parameters of statistical models
- Let's build a simplified model
  - Kansas City has a single winning probability θ for all the games of a season
  - A Kansas City's game can be viewed as a Bernoulli trial with probability θ
- Binomial distribution
  - The probability that Kansas City wins k times out of n games, given the winning probability θ

$$P(k \text{ wins out of } n \text{ games}|\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$$

θ is the parameter to be estimated!

$\binom{n}{k}$ is the number of k-combinations: link

Maximum likelihood estimation (MLE)

- Choose θ that maximizes probability of observed data (likelihood of θ)

$$\theta^* = \arg\max_\theta P(\mathcal{D}|\theta)$$

- If winning probability $\theta = 0.1$, the probability that data D is observed

$$P(12 \text{ wins out of } 17 \text{ games}|\theta = 0.1) = \binom{17}{12} 0.1^{12}(1-0.1)^5 = 3.65 * 10^{-9}$$

- If winning probability $\theta = 0.7$, the probability that data D is observed

$$P(12 \text{ wins out of } 17 \text{ games}|\theta = 0.7) = \binom{17}{12} 0.7^{12}(1-0.7)^5 = 0.21$$

$\theta = 0.7$ gives much higher probability, data D is much more likely to be observed!

θ^* means the optimal value of parameter θ

Maximum likelihood estimation (MLE)

- Choose θ that maximizes probability of observed data (likelihood of θ)

$$\theta^* = \arg\max_\theta P(\mathcal{D}|\theta)$$

- Take derivative of likelihood to search for θ which maximizes $P(\mathcal{D}|\theta)$

$$\frac{dP(D\,|\,\theta)}{d\theta} = \binom{n}{k}\left(k\theta^{k-1}(1-\theta)^{n-k} - (n-k)\theta^k(1-\theta)^{n-k-1}\right)$$

$$= \binom{n}{k}\theta^{k-1}(1-\theta)^{n-k-1}(k(1-\theta) - (n-k)\theta)$$

$$= 0$$

- By solving the equation, θ^* that maximize the likelihood is

$$\boxed{\theta^* = \frac{k}{n}}$$

In our example, $\theta^* = \frac{12}{17} = 71\%$

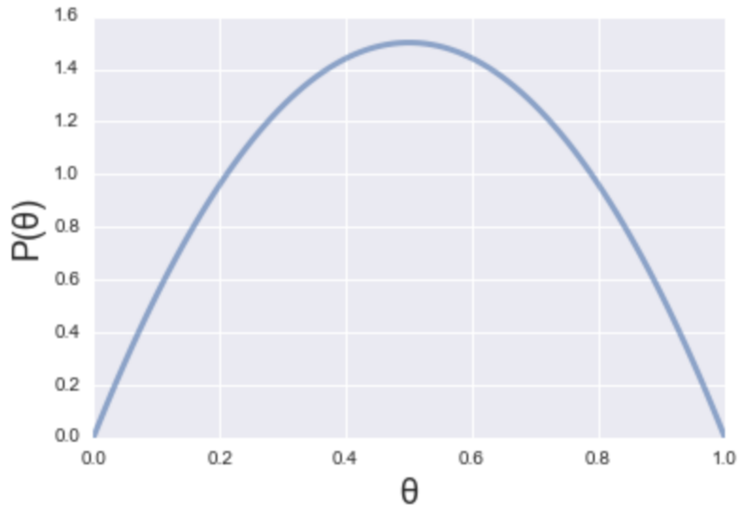
θ^* means the optimal value of parameter θ

Maximum a posteriori (MAP) estimation

- Why do we need MAP?
  - MLE is powerful when you have enough data D (data truthful reflect the probability in real-world)
  - What if data size is small?
  - If Kansas City only had 2 games, does winning probability $\theta = 1$?

Maximum a posteriori (MAP) estimation

- We need to consider prior knowledge besides data
  - Without having data observed, we already knew that θ should be around 0.5
  - We can model $P(\theta)$ in advance as the prior probability of θ



  - Then, having the data D from the last season games, we can update $P(\theta)$ and obtain $P(\theta|\mathcal{D})$, this is **posterior probability**!

Maximum a posteriori (MAP) estimation

- Choose θ that is most probable given prior probability and observed data

$$\theta^* = \arg\max_\theta P(\theta|\mathcal{D})$$

- How to calculate $P(\theta|\mathcal{D})$? We can use Bayes' Rule!

$$\begin{aligned}\theta^* &= \arg\max_\theta \frac{P(\mathcal{D}|\theta)P(\theta)}{P(\mathcal{D})} \\ &= \arg\max_\theta P(\mathcal{D}|\theta)P(\theta)\end{aligned}$$

Since $P(\mathcal{D})$ is independent to the value of θ, we can ignore it in our maximization

θ^* means the optimal value of parameter θ

Maximum a posteriori (MAP) estimation

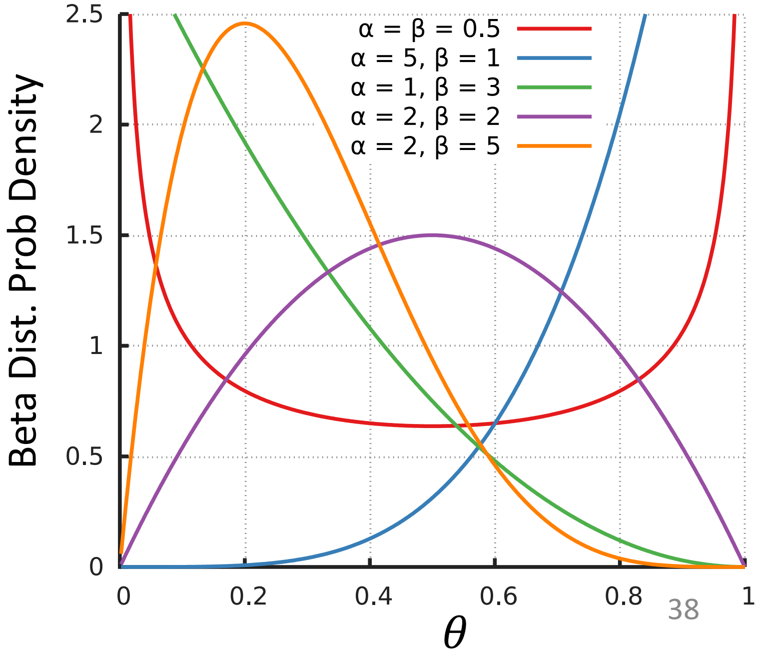
- Since $P(\theta)$ is a belief we have about parameters (winning probability θ)
  - We can use any distribution to describe $P(\theta)$
  - Mostly, we use **conjugate prior distribution**
    - For computational convenience, posterior $P(\theta|\mathcal{D})$ can be expressed as closed-form
    - The prior $P(\theta)$ is in the same probability distribution family with the likelihood $P(\mathcal{D}|\theta)$
  - For the likelihood following binomial distribution

$$P(\mathcal{D}|\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$$

  - We use beta distribution

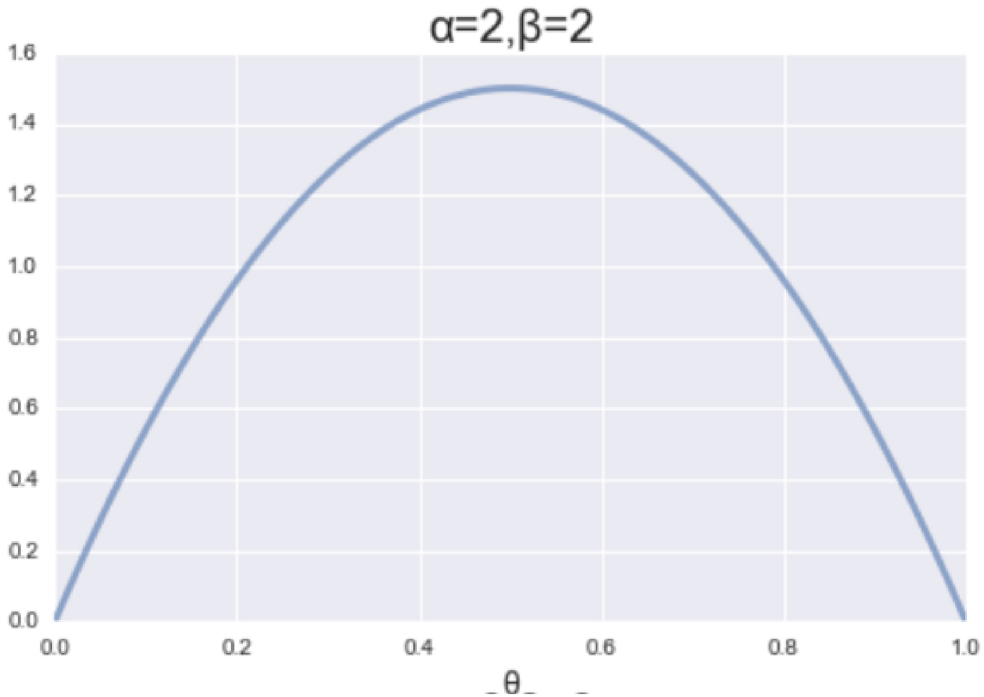
$$P(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$$

α and β are hyperparameters which need to be predefined in advance



Maximum a posteriori (MAP) estimation

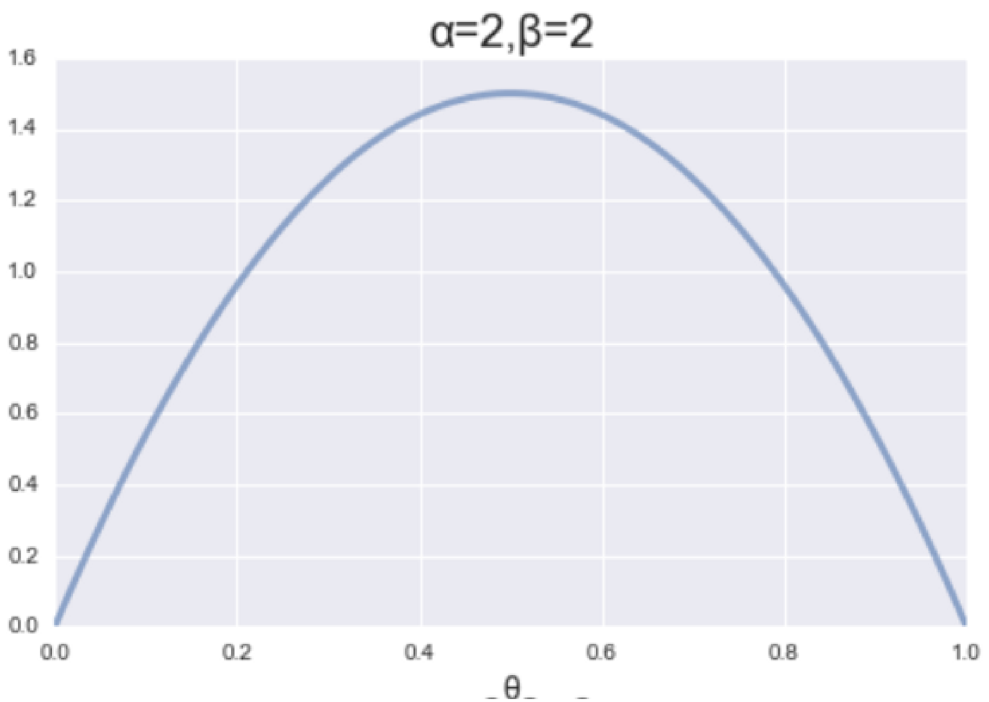
- In our case, we believe that winning probability of Kansas City is
  - Mostly likely to be around 0.5 based on prior knowledge, less likely to be 0 or 1
  - We set $\alpha = 2$ and $\beta = 2$ for beta distribution as prior

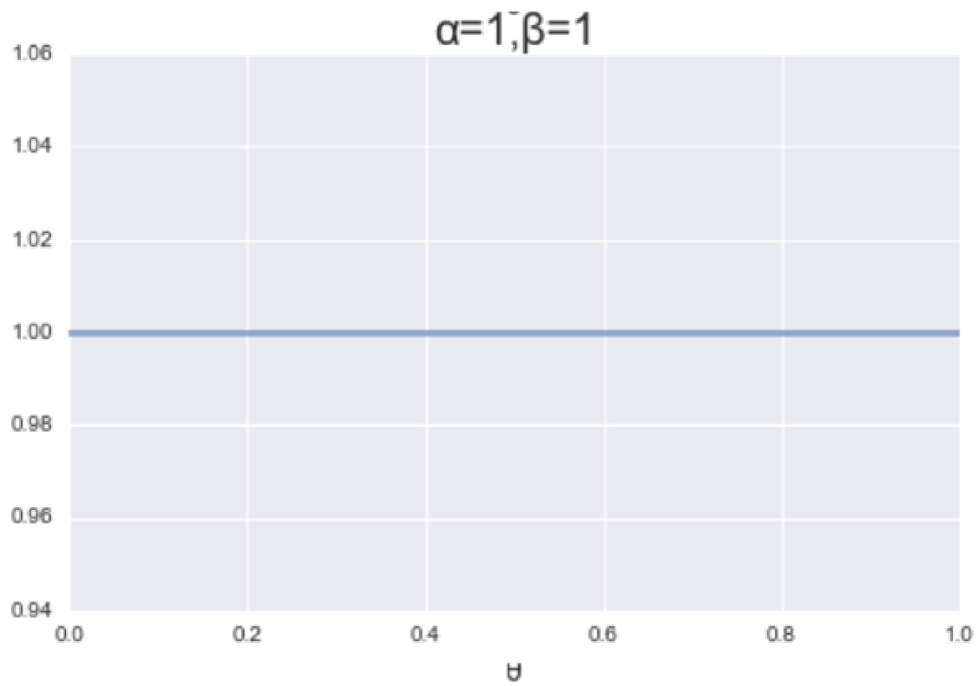


Maximum a posteriori (MAP) estimation

- In our case, we believe that winning probability of Kansas City is
  - Mostly likely to be around 0.5 based on prior knowledge, less likely to be 0 or 1
  - We set $\alpha = 2$ and $\beta = 2$ for beta distribution as prior
  - Question: what happen if we set $\alpha = 1$ and $\beta = 1$?

In this case, we don't have any prior knowledge about θ. MAP will be completely same as MLE





Maximum a posteriori (MAP) estimation

- Choose θ that is most probable given prior probability and observed data

$$\theta^* = \arg\max_\theta P(\mathcal{D}|\theta)P(\theta)$$

- Formulate the posterior distribution $P(\theta|\mathcal{D})$ as

$$P(D\,|\,\theta)P(\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$$

$$= \binom{n}{k}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{k+\alpha-1}(1-\theta)^{n-k+\beta-1}$$

Maximum a posteriori (MAP) estimation

- Choose θ that is most probable given prior probability and observed data

$$\theta^* = \arg\max_{\theta} P(\mathcal{D}|\theta)P(\theta)$$

- Take derivative of posterior $P(\theta|\mathcal{D})$ to search for optimal θ

$$\frac{dP(D\mid\theta)P(\theta)}{d\theta} = \binom{n}{k}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\left((k+\alpha-1)\theta^{k+\alpha-2}(1-\theta)^{n-k+\beta-1} - (n-k+\beta-1)\theta^{k+\alpha-1}(1-\theta)^{n-k+\beta-2}\right)$$

$$= \binom{n}{k}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{k+\alpha-2}(1-\theta)^{n-k+\beta-2}\left((k+\alpha-1)(1-\theta)-(n-k+\beta-1)\theta\right)$$

$$= 0$$

- By solving the equation, θ^* that maximize the posterior is

$$\boxed{\theta^* = \frac{k+\alpha-1}{n+\alpha+\beta-2}}$$

In our example, $\theta^* = \frac{12+2-1}{17+2+2-2} = 68.4\%$

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Bayes classification setting

- A probabilistic framework for solving classification problems
- X, Y are random variables: X represents data and Y represents label
- Bayes' Rule:

$$P(Y = y_k | X = x_i) = \frac{P(X = x_i | Y = y_k) P(Y = y_k)}{P(X = x_i)}$$

$P(Y = y_k | X = x_i)$ means given the data point x_i, the probability of a possible class y_k

- In detail

$$P(Y = y_k | X = x_i) = \frac{P\left(X_1 = x_{i,1} \wedge \cdots \wedge X_d = x_{i,d} \middle| Y = y_k\right) P(Y = y_k)}{P(X_1 = x_{i,1} \wedge \cdots \wedge X_d = x_{i,d})}$$

The probability of a data point $P(X = x_i)$ is the joint probability of all features

Naïve Bayesian classifiers

- Idea: use the training data to estimate $P(X|Y)$ and $P(Y)$
- Then use Bayes rule to infer $P(Y|X_{new})$ for new data
- Example: How to classify the new record X = ('Yes', 'Single', 80K)

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Find the class with the highest probability given the attribute values.

Maximum A Posteriori (MAP) estimate:

- Find the value y for class Y that maximizes P(Y=y| X)

How do we estimate P(Y|X) for the different values of Y?

- We want to estimate P(Y=Yes| X) and P(Y=No| X)

Using Bayes' Rule for classification

- Compute posterior probability $P(Y \mid X_1, X_2, \ldots, X_d)$ using the Bayes Rule

Necessary but impractical

Easy to estimate from data

$$P(Y = y_k | X = x_i) = \frac{\boxed{P\left(X_1 = x_{i,1} \wedge \cdots \wedge X_d = x_{i,d} \middle| Y = y_k\right)}\boxed{P(Y = y_k)}}{\boxed{P(X_1 = x_{i,1} \wedge \cdots \wedge X_d = x_{i,d})}}$$

Unnecessary

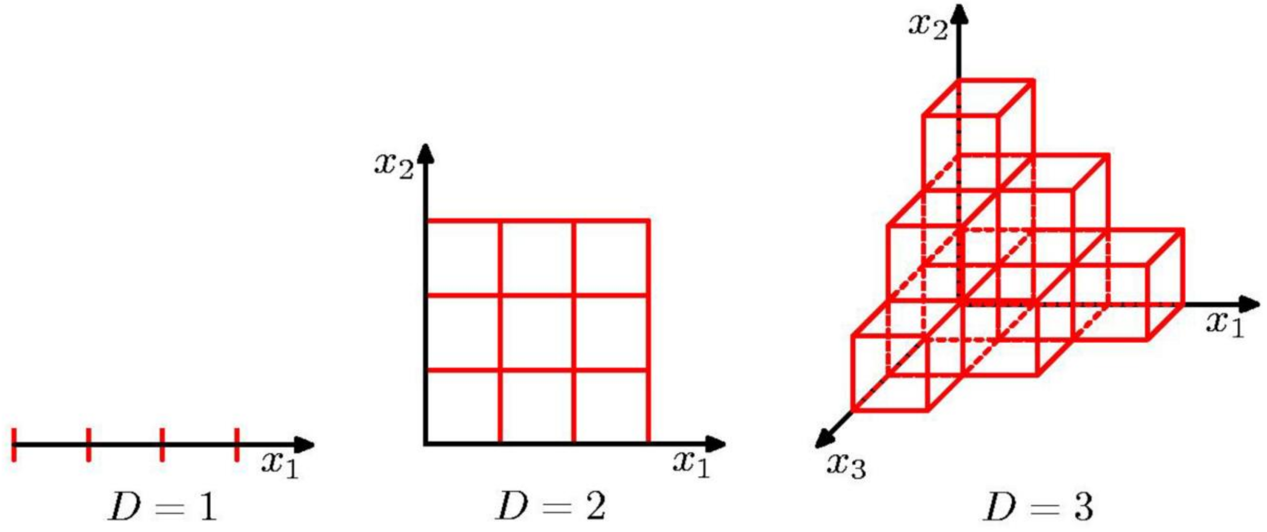
- **Maximum A Posteriori (MAP)**: Choose Y that maximizes

$$P(Y \mid X_1, X_2, \ldots, X_d)$$

- How to estimate class-conditional probability $P(X_1, X_2, \ldots, X_d \mid Y)$?
  - Estimating the joint probability distribution is not practical

Conditional independence

- Compute class-conditional probability $P(X_1, X_2, \ldots, X_d \mid Y)$
  - Is impractical as the number of attribute combinations grow exponentially.
    - E.g., $P(X_1 \mid Y, X_2, \ldots, X_d)$
  - For d boolean attributes, we need the probability of 2^n combinations
  - 30 Boolean attributes need around 1 billion combination (data should be more)
  - Fewer training instances of every combination lead to poor estimation



Conditional independence

- If we make assumption that the attributes are independent given the class label, estimation becomes feasible

$$P(X_1, X_2, ..., X_d|Y) = \prod_{j=1}^{d} P(X_j|Y)$$

- X_1 and X_2 are conditionally independent given Y
  - if $P(X_1 \mid X_2, Y) = P(X_1 \mid Y)$
  - Example: Arm length X_1 and reading skills X_2
    - Young child has shorter arm length and limited reading skills, compared to adults
    - If age Y is fixed, no apparent relationship between arm length and reading skills
    - Arm length and reading skills are conditionally independent given age

Naïve Bayes classifier

- Assume conditional independence among attributes X_i when class Y_j is given:
  - $P(X_1, X_2, \ldots, X_d | Y_j) = P(X_1 | Y_j)\, P(X_2 | Y_j) \ldots P(X_d | Y_j)$

  - Now we can estimate $P(X_i | Y_j)$ for all X_i and Y_j combinations from the training data

  - New point $X = (X_1, \ldots, X_n)$ is classified to class y if

$$P(Y = y|X) = P(Y = y) \prod_i P(X_i|y)$$

is maximum over all possible values of Y.

Example

Record

- X = (Refund = Yes, Status = Single, Income =80K)

- For the class Y = 'Evade', we want to compute:

  P(Y = Yes|X) and P(Y = No|X)

- We compute:
  - P(Y = Yes|X) = P(Y = Yes)*P(Refund = Yes |Y = Yes)
    *P(Status = Single |Y = Yes)
    *P(Income =80K |Y= Yes)
  - P(Y = No|X) = P(Y = No)*P(Refund = Yes |Y = No)
    *P(Status = Single |Y = No)
    *P(Income =80K |Y= No)

How to estimate probabilities from data?

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Class Prior Probability:

$$P(Y = y) = \frac{N_y}{N}$$

N_y: Number of records with class y

N: Number of records

P(Y = Yes) = 3/10

P(Y = No) = 7/10

How to estimate probabilities from data?

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
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3	No	Single	70K	No
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Discrete attributes:

$$P(X_i = x|Y = y) = \frac{N_{x,y}}{N_y}$$

$N_{x,y}$: number of instances having attribute $X_i = x$ and belong to class y

N_y: number of instances of class y

How to estimate probabilities from data?

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
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Discrete attributes:

$$P(X_i = x | Y = y) = \frac{N_{x,y}}{N_y}$$

$N_{x,y}$: number of instances having attribute $X_i = x$ and belong to class y

N_y: number of instances of class y

P(Refund = Yes|No) = 3/7

How to estimate probabilities from data?

Tid	**Refund**	**Marital Status**	**Taxable Income**	**Evade**
1	Yes	Single	125K	No
2	No	Married	100K	No
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Discrete attributes:

$$P(X_i = x | Y = y) = \frac{N_{x,y}}{N_y}$$

$N_{x,y}$: number of instances having attribute $X_i = x$ and belong to class y

N_y: number of instances of class y

P(Refund = Yes|Yes) = 0

How to estimate probabilities from data?

Tid	**Refund**	**Marital Status**	**Taxable Income**	**Evade**
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
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Discrete attributes:

$$P(X_i = x | Y = y) = \frac{N_{x,y}}{N_y}$$

$N_{x,y}$: number of instances having attribute $X_i = x$ and belong to class y

N_y: number of instances of class y

P(Status = Single|No) = 2/7

How to estimate probabilities from data?

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
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8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Discrete attributes:

$$P(X_i = x|Y = y) = \frac{N_{x,y}}{N_y}$$

$N_{x,y}$: number of instances having attribute $X_i = x$ and belong to class y

N_y: number of instances of class y

P(Status = Single|Yes) = 2/3

How to estimate probabilities from data?

- For continuous attributes:
  – Discretization: Partition the range into bins:
    - Replace continuous value with bin value
      - Attribute changed from continuous to ordinal

  – Probability density estimation:
    - Assume attribute follows a distribution (e.g., normal distribution)
    - Use data to estimate parameters of distribution through MLE (e.g., mean and standard deviation)
      - The MLE for mean and variance is computed by $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$
    - Once probability distribution is known, use it to estimate the conditional probability $P(X_i|Y)$

MLE for Normal distribution:
https://towardsdatascience.com/maximum-likelihood-estimation-explained-normal-distribution-6207b322e47f

How to estimate probabilities from data?

Tid	Refund	Marital Status	Taxable Income	Evade
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Normal distribution:

$$P(X_i = x | Y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(a-\mu_{ij})^2}{2\sigma_{ij}^2}}$$

  - One for each (X_i, Y_j) pair

- For Class=No
  - sample mean μ = 110
  - sample variance σ^2= 2975
- For Income = 80

$$P(\text{Income} = 80|\text{No}) = \frac{1}{\sqrt{2\pi 2975}} e^{-\frac{(80-110)^2}{2(2975)}} = 0.0062$$

Sample variance calculation: https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1008&context=imseteach

How to estimate probabilities from data?

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Normal distribution:

$$P(X_i = x \mid Y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(a-\mu_{ij})^2}{2\sigma_{ij}^2}}$$

  - One for each (X_i, Y_j) pair

- For Class=Yes
  - sample mean $\mu = 90$
  - sample variance $\sigma^2 = 25$
- For Income = 80

$$P(\text{Income} = 80 | \text{Yes}) = \frac{1}{\sqrt{2\pi 25}} e^{-\frac{(80-90)^2}{2(25)}} = 0.0108$$

Example

Record

- X = (Refund = Yes, Status = Single, Income =80K)

- For the class Y = ‘Evade’, we want to compute:

  P(Y = Yes|X) and P(Y = No|X)

- We compute:
  - P(Y = Yes|X) = P(Y = Yes)*P(Refund = Yes |Y = Yes)
    *P(Status = Single |Y = Yes)
    *P(Income =80K |Y= Yes)
    = 3/10* 0 * 2/3 * 0.01 = 0
  - P(Y = No|X) = P(Y = No)*P(Refund = Yes |Y = No)
    *P(Status = Single |Y = No)
    *P(Income =80K |Y= No)
    = 7/10 * 3/7 * 2/7 * 0.0062 = 0.0005

Example of Naïve Bayes classifier

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$$

Naïve Bayes Classifier:

P(Refund = Yes | No) = 3/7
P(Refund = No | No) = 4/7
P(Refund = Yes | Yes) = 0
P(Refund = No | Yes) = 1
P(Marital Status = Single | No) = 2/7
P(Marital Status = Divorced | No) = 1/7
P(Marital Status = Married | No) = 4/7
P(Marital Status = Single | Yes) = 2/3
P(Marital Status = Divorced | Yes) = 1/3
P(Marital Status = Married | Yes) = 0

For Taxable Income:
If class = No: sample mean = 110
sample variance = 2975
If class = Yes: sample mean = 90
sample variance = 25

- P(X | No) = P(Refund=No | No) $\times$ P(Divorced | No) $\times$ P(Income=120K | No) = 4/7 $\times$ 1/7 $\times$ 0.0072 = 0.0006
- P(X | Yes) = P(Refund=No | Yes) $\times$ P(Divorced | Yes) $\times$ P(Income=120K | Yes) = 1 $\times$ 1/3 $\times$ 1.2 $\times$ 10^{-9} = 4 $\times$ 10^{-10}

Since P(X|No)P(No) > P(X|Yes)P(Yes)

Therefore P(No|X) > P(Yes|X)
=> Class = No

Naïve Bayes Classifier performs under missing attributes

Even in absence of information about any attributes, we can use A priori Probabilities of Class Variable:

P(Yes) = 3/10

P(No) = 7/10

Naïve Bayes Classifier:

P(Refund = Yes | No) = 3/7
P(Refund = No | No) = 4/7
P(Refund = Yes | Yes) = 0
P(Refund = No | Yes) = 1
P(Marital Status = Single | No) = 2/7
P(Marital Status = Divorced | No) = 1/7
P(Marital Status = Married | No) = 4/7
P(Marital Status = Single | Yes) = 2/3
P(Marital Status = Divorced | Yes) = 1/3
P(Marital Status = Married | Yes) = 0

For Taxable Income:
If class = No: sample mean = 110
sample variance = 2975
If class = Yes: sample mean = 90
sample variance = 25

If we only know that marital status is Divorced, then:

P(Yes | Divorced) = 1/3 x 3/10 / P(Divorced)

P(No | Divorced) = 1/7 x 7/10 / P(Divorced)

If we also know that Refund = No, then

P(Yes | Refund = No, Divorced) = 1 x 1/3 x 3/10 / P(Divorced, Refund = No)

P(No | Refund = No, Divorced) = 4/7 x 1/7 x 7/10 / P(Divorced, Refund = No)

If we also know that Taxable Income = 120, then

P(Yes | Refund = No, Divorced, Income = 120) = 1.2 x10^{-9} x 1 x 1/3 x 3/10 / P(Divorced, Refund = No, Income = 120)

P(No | Refund = No, Divorced Income = 120) = 0.0072 x 4/7 x 1/7 x 7/10 / P(Divorced, Refund = No, Income = 120)

Issues with Naïve Bayes classifier

Consider the table with Tid = 7 deleted

Tid	**Refund**	**Marital Status**	**Taxable Income**	**Evade**
1	Yes	Single	125K	**No**
2	No	Married	100K	**No**
3	No	Single	70K	**No**
4	Yes	Married	120K	**No**
5	No	Divorced	95K	**Yes**
6	No	Married	60K	**No**
~~7~~	~~Yes~~	~~Divorced~~	~~220K~~	~~**No**~~
8	No	Single	85K	**Yes**
9	No	Married	75K	**No**
10	No	Single	90K	**Yes**

Naïve Bayes Classifier:

P(Refund = Yes | No) = 2/6
P(Refund = No | No) = 4/6
P(Refund = Yes | Yes) = 0
P(Refund = No | Yes) = 1
P(Marital Status = Single | No) = 2/6
P(Marital Status = Divorced | No) = 0
P(Marital Status = Married | No) = 4/6
P(Marital Status = Single | Yes) = 2/3
P(Marital Status = Divorced | Yes) = 1/3
P(Marital Status = Married | Yes) = 0/3
For Taxable Income:
If class = No: sample mean = 91
sample variance = 685
If class = No: sample mean = 90
sample variance = 25

Given X = (Refund = Yes, Divorced, 120K)

P(X | No) = 2/6 X 0 X 0.0083 = 0

P(X | Yes) = 0 X 1/3 X 1.2 X 10^{-9} = 0

Naïve Bayes will not be able to classify X as Yes or No!

Laplace Smoothing

- If one of the conditional probabilities is zero, then the entire expression becomes zero

- Laplace Smoothing:

$$P(A_i = a|C = c) = \frac{N_{ac} + 1}{N_c + N_i}$$

  - N_i: number of unique attribute values for attribute A_i

Example of Laplace Smoothing

- Creating a Naïve Bayes Classifier, essentially means to compute counts:

With Laplace Smoothing

Total number of records: N = 10

Class No:
Number of records: 7
Attribute Refund:
Yes: 3
No: 4
Attribute Marital Status:
Single: 2
Divorced: 1
Married: 4
Attribute Income:
mean: 110
variance: 2975

Class Yes:
Number of records: 3
Attribute Refund:
Yes: 0
No: 3
Attribute Marital Status:
Single: 2
Divorced: 1
Married: 0
Attribute Income:
mean: 90
variance: 25

naive Bayes Classifier:

P(Refund=Yes|No) = 4/9
P(Refund=No|No) = 5/9
P(Refund=Yes|Yes) = 1/5 (0+1)/(3+2)
P(Refund=No|Yes) = 4/5

P(Marital Status=Single|No) = 3/10
P(Marital Status=Divorced|No)=2/10
P(Marital Status=Married|No) = 5/10
P(Marital Status=Single|Yes) = 3/6
P(Marital Status=Divorced|Yes)=2/6
P(Marital Status=Married|Yes) = 1/6
(0+1)/(3+3)

For taxable income:
If class=No: sample mean=110
sample variance=2975
If class=Yes: sample mean=90
sample variance=25

Example of Laplace Smoothing

Given a Test Record:

With Laplace Smoothing

X = (Refund = Yes, Status = Single, Income =80K)

naive Bayes Classifier:

P(Refund=Yes|No) = 4/9
P(Refund=No|No) = 5/9
P(Refund=Yes|Yes) = 1/5
P(Refund=No|Yes) = 4/5

P(Marital Status=Single|No) = 3/10
P(Marital Status=Divorced|No)=2/10
P(Marital Status=Married|No) = 5/10
P(Marital Status=Single|Yes) = 3/6
P(Marital Status=Divorced|Yes)=2/6
P(Marital Status=Married|Yes) = 1/6

For taxable income:
If class=No: sample mean=110
sample variance=2975
If class=Yes: sample mean=90
sample variance=25

P(X|Class=No) = P(Refund=No|Class=No)
× P(Married| Class=No)
× P(Income=120K| Class=No)
= 4/9 × 3/10 × 0.0062 = 0.00082

P(X|Class=Yes) = P(Refund=No| Class=Yes)
× P(Married| Class=Yes)
× P(Income=120K| Class=Yes)
= 1/5 × 3/6 × 0.01 = 0.001

- P(No) = 0.7, P(Yes) = 0.3
- P(X|No)P(No) = 0.0005
- P(X|Yes)P(Yes) = 0.0003

=> Class = No

Naïve Bayes for text classification

- Naïve Bayes is commonly used for text classification
- For a document with k terms $d = (t_1, \ldots, t_k)$

$$P(c|d) = P(c)P(d|c) = P(c)\prod_{t_i \in d} P(t_i|c)$$

Fraction of documents in class c

- $P(t_i|c)$= Fraction of terms from all documents in c that are t_i.

Number of times t_i appears in all documents in c

$$P(t_i|c) = \frac{N_{ic} + 1}{N_c + T}$$

Laplace Smoothing

Number of unique words (vocabulary size)

Total number of terms in all documents in c

- Easy to implement and works relatively well
- Limitation: Hard to incorporate additional features (beyond words).
  - E.g., number of adjectives used.

Example

News titles for Politics and Sports

	Politics	Sports	
documents	"Obama meets Merkel" "Obama elected again" "Merkel visits Greece again"	"OSFP European basketball champion" "Miami NBA basketball champion" "Greece basketball coach?"	Vocabulary size: 14
	P(p) = 0.5	P(s) = 0.5	
terms	obama:2, meets:1, merkel:2, elected:1, again:2, visits:1, greece:1	OSFP:1, european:1, basketball:3, champion:2, miami:1, nba:1, greece:1, coach:1	
	Total terms: 10	Total terms: 11	

New title: X = "Obama likes basketball"

P(Politics|X) ~ P(p)*P(obama|p)*P(likes|p)*P(basketball|p)
= 0.5 * 3/(10+14) *1/(10+14) * 1/(10+14) = 0.000108

P(Sports|X) ~ P(s)*P(obama|s)*P(likes|s)*P(basketball|s)
= 0.5 * 1/(11+14) *1/(11+14) * 4/(11+14) = 0.000128

Naïve Bayes summary

- A generative model by treating target class as causative factor for generating the data instance – descriptive insights
- Robust to isolated noise points –averaged out in conditional probability
- Handle missing values by ignoring the instance during probability estimate calculations
- Robust to irrelevant attributes as P(X|Y) uniformly distributed
- Correlated attributes can degrade the model - independence assumption may not hold
  - Use other techniques such as Bayesian Networks
- Naïve Bayes can produce a probability estimate, but it is usually very biased
  - Logistic Regression is better for obtaining probabilities.